e and Natural Logarithms

Questions 1–46

Question 1. Solve the following equations, writing your answer in logarithmic form:

a) $e^x = 8$
b) $e^{2x} = 4$
c) $e^{3x} = 8$
d) $e^{5x} = \frac{1}{100}$

Question 2. Solve each of these equations:

a) $e^{4x+3} = 9$
b) $e^{5x-2} = 10$
c) $e^{9-2x} = 4$
d) $e^{1-5x} = \frac{1}{5}$

Question 3. Solve the following equations, writing your answer in the form $x = e^a$.

a) $\ln x = 7$
b) $\ln x = -5$
c) $\ln x = \frac{1}{3}$
d) $\ln x = -\frac{2}{5}$

Question 4. Write each of these as a single logarithm:

a) $\ln 4 + \ln 5 + \ln 6$
b) $\ln 8 + \ln 3 - \ln 4$
c) $\ln 12 - \ln 4 - \ln 6$
d) $\ln 40 - \ln 8 + \ln 5$

Question 5. Solve the following equations:

a) $\ln x + \ln 2 = \ln 8$
b) $\ln x - \ln 5 = \ln 3$
c) $\ln 8 + \ln x = \ln \frac{1}{3}$
d) $\ln 50 - \ln x = \ln \frac{3}{2}$

Question 6. Evaluate each of these:

a) $\ln e$
b) $4 \ln \frac{1}{e}$
c) $\ln \frac{1}{e^5}$
d) $\ln \sqrt[3]{e}$

Question 7. Write each of these as a single logarithm:

a) $2\ln 2 + 2\ln 4$

b) $3\ln 2 - 2\ln 3$

c) $2\ln 9 + 3\ln 3 - 2\ln 6$

d) $2\ln 10 - 4\ln 2 - 2\ln 5$

Question 8. Write $3\ln x - 4\ln y$ as a single logarithm.

Question 9. Write each of these as a single logarithm:

a) $4\ln p + 2\ln q + 3\ln r$
b) $3\ln p + 5\ln q - 4\ln r$
c) $6\ln p - 2\ln q - 8\ln r$
d) $9\ln p - 5\ln q + 7\ln r$

Question 10. Write each of these as a single logarithm:

a) $2\ln p - (\ln q + 3\ln r)$
b) $\ln p - (2\ln q - \ln r)$
c) $3\ln p - 2(\ln q + 2\ln r)$
d) $4\ln p - 3(2\ln q - 3\ln r)$

Question 11. Write each of these equations in a form not involving logarithms:

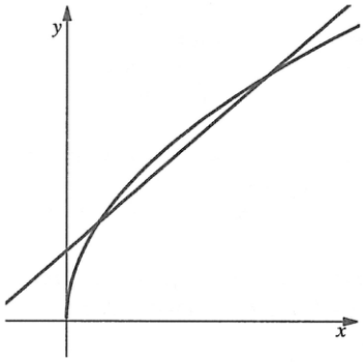
a) $\ln x + \ln y = 5$

b) $\ln y - \ln x = 12$

c) $2\ln x - \ln y = 4$

d) $4\ln x + 3\ln y = 8$

Question 12. The diagram below shows the straight-line graph $7y = x + 10$ and the curve with equation $\ln x = 2\ln y$. Determine the exact coordinates of where the two graphs intersect.



Question 13. Solve each of these equations:

a) $e^{2x} - 16e^{x} = 0$

b) $e^{3x} - 16e^{x} = 0$

c) $e^{3x} - 16e^{2x} = 0$

d) $e^{4x} - 16e^{2x} = 0$

Question 14. Solve each of these equations:

a) $e^{x} = e^{2x+1}$

b) $e^{2x} = e^{3x-5}$

c) $e^{x} = 2e^{4x}$

d) $e^{2x} = 3e^{x-1}$

Question 15.

a) Using $y = e^x$ as a substitution, show that

$$4e^x + 27e^{-x} - 39 = 0$$

can be written as

$$4y^2 - 39y + 27 = 0$$

b) Hence solve the equation

$$4e^x + 27e^{-x} - 39 = 0$$

giving your answers as exact values.

Question 16. Solve the following equations, writing your answers in logarithmic form:

a) $e^{2x} - 6e^x + 8 = 0$

b) $e^{2x} - 13e^x + 40 = 0$

c) $e^x + 4 - 45e^{-x} = 0$

d) $e^x + 5 - 150e^{-x} = 0$

Question 17. Solve the following equations, writing your answers in logarithmic form:

a) $2e^{2x} - 7e^{x} + 6 = 0$

b) $4e^{2x} - 28e^{x} + 45 = 0$

c) $6e^{x} - 23 - 18e^{-x} = 0$

d) $8e^{x} + 14 - 15e^{-x} = 0$

Question 18. Show that the solution to the equation

$$4^{x-3} = 5^{x+6}$$

can be written in the form

$$\frac{6\ln 5 + 3\ln 4}{\ln 4 - \ln 5}$$

Question 19. By writing $\ln 5 \times \ln 125 \times \ln 625$ in terms of $\ln 5$, show that

$$\ln 5 \times \ln 125 \times \ln 625 > 12$$

Question 20. Solve each of these equations by taking natural logarithms of both sides:

a) $2^x = 5^{x+4}$
b) $4^{x-1} = 6^{x+2}$
c) $5^{3-x} = 7^{x+1}$
d) $8^{9-x} = 6^{8-x}$

Question 21. Solve each of these equations, writing your answers in exact form:

a) $2\ln(3x-5)-4=0$

b) $2\ln(3-2x)-9=0$

c) $2\ln(x+4)-\ln 5=0$

d) $5\ln 2x-3\ln 3x=1$

Question 22. Solve the equation $2\ln x-\ln(x+12)=0$

Question 23. Solve each of these equations, writing your answers in exact form:

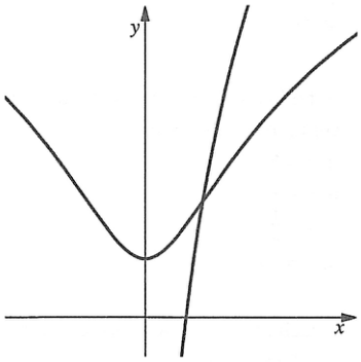
a) $\ln x = 2 + \ln(x + 3)$

b) $\ln(x - 1) = 3 + \ln(x + 1)$

c) $\ln 2x = 4 - \ln x$

d) $\ln 4x = 5 - \ln(x + 1)$

Question 24. The diagram below shows the curves $y = \ln(x^2 + 2)$ and $y = 4 \ln x$. Find the exact coordinates of where the two curves intersect.



Question 25. Solve each of these equations, writing your answers in exact form:

a) $2\ln x = 2 + \ln 2x$

b) $2\ln x = 3 - \ln 8x$

c) $2\ln x = \ln 4x - 1$

d) $2\ln x = \ln 10x - 4$

Question 26. Solve the equation

$$3\ln x = 3 - \ln 16x$$

writing your answer in exact form.

Question 27. Solve each of these equations, writing your answers in exact form:

a) $\ln x + \frac{32}{\ln x} = 12$

b) $2\ln x + \frac{6}{\ln x} = 7$

c) $6\ln x + 11 = \frac{10}{\ln x}$

d) $10\ln x - 33 = \frac{7}{\ln x}$

Question 28. Solve the equation

$$2\ln(x+10) = \ln x + \ln(2x-1)$$

Question 29. Solve the equation

$$\ln(5-3x)+\ln(8+4x)=2\ln(x+2)$$

Question 30. Solve the equation

$$2\ln(5-2x)=\ln 2+\ln(\frac{233}{8}-2(x-\frac{5}{4})^2)$$

Question 31.

a) Using $y = e^x$ as a substitution, show that

$$1 - 28e^{-3x} - 4e^{-2x} + 7e^{-x} = 0$$

can be written as

$$y^3 + 7y^2 - 4y - 28 = 0$$

b) Hence solve the equation

$$1 - 28e^{-3x} - 4e^{-2x} + 7e^{-x} = 0$$

giving your answers as exact values.

Question 32. Given that $y = \ln x$ and $y^2 - 3y + 2 = 0$, find the possible values of x

Question 33. Solve the equation

$$6(\ln x)^2 - 19\ln x + 15 = 0$$

Question 34. Solve the equation

$$\ln 2x + \ln 2x^2 + \ln 2x^3 = 6$$

Question 35. Solve the equation

$$\frac{e^x}{3^x} = e$$

writing your answer in the form

$$\frac{p}{q + \ln r}$$

where p, q and r are integers.

Question 36. Solve the equation

$$3^x e^{2x+3} = 2$$

writing your answer in the form

$$\frac{p + \ln q}{r + \ln s}$$

where p, q, r and s are integers.

Question 37. Solve the equation

$$5^x e^{8x} = e^{10}$$

writing your answer in the form

$$\frac{p}{q + \ln r}$$

where p, q and r are integers.

Question 38. Fully simplify

$$\frac{3x^2 - 19x + 6}{x^2 - x - 30}$$

Hence, given that

$$\ln(3x^2 - 19x + 6) - 1 = \ln(x^2 - x - 30)$$

where $x \neq 6$, find x in terms of e.

Question 39. Determine the exact coordinates of where the curves $y = 2e^x - 11$ and $y = 40e^{-x}$ intersect.

Question 40. A curve has equation $y = ae^x + b$ and passes through the points $(\ln 2, 12)$ and $(-3\ln 2, \frac{87}{16})$. Find the values of a and b

Question 41. A curve has equation $y = p\ln x + q$ and passes through the points $(e^{12}, -\frac{55}{2})$ and $(e, \frac{11}{6})$. Find the values of p and q

Question 42. Solve the simultaneous equations

$$e^x + y^2 = 4$$

$$2\ln y = x$$

Question 43. Determine the exact coordinates of where the curves $x + e^y = 4$ and $\ln((x-4)^2) - y = 2$ intersect.

Question 44. The water level, h m, in a large tank over time, t hours, is modelled by the equation

$$h = 20 - 5\ln(t+1)$$

a) What is the initial height of the water in the tank?
b) How high is the water in the tank after 1 hour?
c) After how many hours will the water level be 10 metres?

Question 45. A population of hawks, P, can be modelled by the equation

$$P = 100\ln(\frac{t+3}{3}) + 500$$

where t is the number of years since a conservation project started.

a) What was the initial population of hawks when the conservation project started?

b) What does the model suggest the size of the population will be in 5 years after the conservation project started?

c) After how many years does the model suggest the population will reach 700 hawks?

Question 46. The temperature, T°C, of the water in a lake can be modelled by the equation

$$T = 2\ln(t+3) - \ln(t+2) + \ln 100$$

where t is the number of days after the 1st March.

a) What was the temperature of the lake on the 1st March?

b) How many days after the 1st March will the temperature of the lake be $(\ln 3000)$°C?